



Checking understanding of factors, multiples, squares and cubes

Task A: Multiples

- 1) Using the grid:
- a) Draw a circle next to all the **multiples** of 2
- b) Draw a square next to all the **multiples** of 3
- c) Draw a triangle next to all the **multiples** of 5
- d) What do the numbers with a circle and square represent?

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35

Multiples of 2 and 3

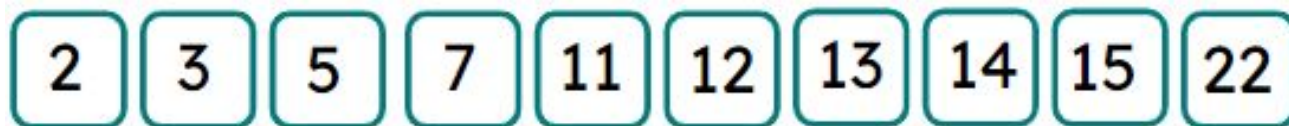
- e) What do the numbers with a circle, square and triangle represent?

Multiples of 2, 3 and 5



Task B: Factors

1) Only using the following cards once, fill in the gaps.



a) **14** is a multiple of **7**

d) **11** is a factor of **22**

b) **12** is a multiple of **3**

e) **13** only has **2** factors

c) **15** is a multiple of **5**

2) Are the following always true, sometimes true or never true? When the answer is sometimes or never true, give an example

	Always true	Sometimes true	Never true
Given A, B and C are integers $A \times B = C$ C is a multiple of A and B A and B are factors of C	✓		
A multiple of a number is bigger than its largest factor		<i>5 is a multiple of 5 and 5 is a factor of 5</i>	
A product of two integers is an integer	✓		
1 is a factor of every integer	✓		



Task C: Square and cube numbers

1) Work out the unknown number given the clues.

- a)

I am less than 100
I am greater than 50
I am a square number
I am odd

81

b)

I am a cube number
The sum of my digits is 9
I am a multiple of 9
I am less than 100

27

c)

I am a factor of 24
I am a multiple of 2
I am a number less than 10
I am a square number

4

d)

I am a cube number
I am a square number
I am a factor of every number

1

2) Complete the Venn diagram showing all the square and cube numbers up to 200.

